ECOSETU

Green compliance, bridged.

Principle 6 Calculation Methodology

How Ecosetu turns simple operational inputs - electricity bills, diesel litres, employee counts - into BRSR-grade environmental disclosures, and the sources behind every conversion.

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Intended audience:
Chartered Accountants, ESG consultants, advocates and other professional reviewers who advise MSME clients on BRSR-style sustainability reporting.

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v1.1 update: Scope 2 grid factor migrated to CEA V21.0 (Nov 2025)

1. Why this document exists

When a CA or ESG consultant first sees Ecosetu, the same question comes up almost every time: "How are you generating Scope 1 and Scope 2 emissions from just an electricity bill and a diesel figure? How is water usage being estimated from headcount? Can I trust these numbers enough to put them in front of a client - or worse, a regulator?"

That is the right question, and this document is the honest answer.

Ecosetu is not magic. It is a thin layer of well-established environmental engineering and accounting conventions, wrapped in a workflow that an Indian MSME finance head or compliance manager can actually complete in under an hour. Every number the tool produces follows from one of three things:

- A direct user input (e.g. 12,000 units of electricity per month).
- A standard published conversion factor (e.g. India's grid emission factor, kWh to GJ).
- A widely-used MSME benchmark when ground data is unavailable (e.g. 40 litres per person per day for office water consumption).

No machine learning. No black box. No proprietary "score" hiding behind the disclosures. Every value can be reproduced on the back of an envelope, and this document gives you exactly that envelope.

If you find a factor or assumption you would change, please tell me. I would rather get a correction from a domain expert than ship the wrong number into a thousand reports.

2. The principle behind the design

MSMEs rarely have audited data. They almost always have bills.

A listed company can hand you a six-page energy register with hourly meter readings, fuel logs signed by the maintenance head, and water consumption split by process line. An MSME typically cannot. What an MSME does have, almost without exception:

- Twelve electricity bills a year from the DISCOM.
- A rough number of diesel litres consumed in the DG set.
- A headcount and a working-days figure.
- A rough idea of how many water tankers they buy or borewell hours they run.

Ecosetu starts from those four anchors. From there, every disclosure - energy in GJ, Scope 1, Scope 2, water in kL, intensities per rupee of turnover - flows through documented conversion factors. The tool is built so that the numbers are defensible, the methodology is transparent, and the user does not need a CSR officer to operate it.

Estimation is flagged, never hidden.

Wherever Ecosetu estimates a figure rather than reading a direct measurement, it says so on screen with an estimation flag and shows the underlying formula. The user always sees the difference between what they entered and what the tool inferred. This is essential for audit defensibility - the reviewer can always trace any number back to the input or the assumption.

3. All conversion factors used in Principle 6

Every number Ecosetu produces in Principle 6 ultimately comes from one of the constants below. These are hardcoded at the top of the calculation module, transparent in the public source code, and grounded in the references listed in Section 8.

Constant	Value	Used for
Electricity rate per unit	Rs 8.0 / kWh	Estimating kWh from bill amount
kWh -> Megajoule (MJ)	x 3.6	Energy unit conversion
kWh -> Gigajoule (GJ)	x 0.0036	BRSR-grade energy disclosure
Diesel energy content	9.9 kWh / litre	Energy from DG set fuel
Diesel emission factor	2.68 kgCO2e / L	Scope 1 from diesel combustion
Grid emission factor	0.71 kgCO2e / kWh	Scope 2 from purchased electricity
LPG emission factor	2.98 kgCO2e / kg	Scope 1 from LPG (kitchens, ovens)
Tanker volume	5,000 litres	Tanker count -> water in kL
Litres -> Kilolitres (kL)	x 0.001	Standardising water disclosures
Per-capita office water	40 L / person / day	Estimating water from headcount

These ten values are the entire numerical engine of Principle 6. If any of them moves, the disclosures move predictably and proportionally - no hidden adjustments.

v1.1 update (May 2026): The grid emission factor was updated from 0.82 to 0.71 kgCO2e/kWh following migration to CEA CO2 Baseline Database Version 21.0 (November 2025). See Section 8 for full citation.

4. Energy: kWh, MJ, GJ and intensities

4.1 Estimating electricity units from a bill amount

If a user does not know their kWh consumption but does know the monthly electricity bill in rupees, the tool estimates units consumed using a standard MSME tariff.

```
Estimated kWh per month = Monthly bill (Rs) / Rs 8.0 per unit
```

Worked example

An MSME pays an average of Rs 96,000 per month for electricity. Estimated consumption = 96,000 / 8 = 12,000 kWh per month, or 144,000 kWh per year.

The Rs 8 per unit figure is a conservative all-in commercial tariff. Real tariffs vary across states and slabs (typically Rs 6.50 to Rs 9.50 for industrial users). When the user enters a known kWh figure directly, this estimator is bypassed completely.

4.2 Converting kWh to GJ (the BRSR reporting unit)

BRSR requires total energy to be reported in Gigajoules. The conversion is a fixed physical constant.

```
Energy (GJ) = kWh x 0.0036
```

Worked example

144,000 kWh per year x 0.0036 = 518.4 GJ per year. This is what appears on the BRSR Section C, Principle 6, parameter 1.1.b.

4.3 Adding diesel to total energy

Diesel generators contribute meaningfully to energy and emissions for many manufacturing MSMEs. Diesel litres are converted to energy via its calorific value.

```
Energy from diesel (kWh) = Diesel litres x 9.9
This is then added to the electricity kWh before converting to GJ.
```

Worked example

8,000 litres of diesel per year x 9.9 = 79,200 kWh equivalent. Combined with electricity: (144,000 + 79,200) x 0.0036 = 803.5 GJ total energy.

4.4 Energy intensity per rupee of turnover

BRSR requires intensity ratios so that companies of different sizes can be compared. Ecosetu pulls turnover from Section A and computes the ratio automatically.

```
Energy intensity = Total energy (GJ) / Turnover (Rs)
```

Turnover is taken in absolute rupees, not lakhs or crores, so the resulting figure is a small decimal - this matches how BRSR Section C presents intensity ratios.

5. Emissions: Scope 1 and Scope 2

This is the section that most often draws questions, so it gets the most detail. The principles are identical to what the GHG Protocol and CDP have published for over a decade - Ecosetu is not inventing methodology, it is applying it consistently to MSME-friendly inputs.

5.1 Scope 1: direct emissions you control

Scope 1 covers fuel burned on the company's own premises. For most Indian MSMEs this means two sources: diesel for the DG set, and LPG for canteen / process heating.

Scope 1 from diesel

```
Scope 1 (tCO2e) = Diesel litres x 2.68 kg CO2e / litre / 1000
```

Worked example

8,000 litres of diesel x 2.68 = 21,440 kg CO_2e = 21.44 tCO_2e per year. This number appears directly in BRSR Section C, Principle 6, parameter 2.1.a.

The 2.68 kg/litre factor comes from the IPCC 2006 Guidelines for National Greenhouse Gas Inventories, Volume 2 (Energy), and matches the figure used by CEA and the Bureau of Energy Efficiency in India.

Scope 1 from LPG

```
Scope 1 (tCO2e) = LPG kg x 2.98 kg CO2e / kg / 1000
```

Applies if the user enables the LPG input - many MSMEs do not have material LPG consumption.

5.2 Scope 2: indirect emissions from purchased electricity

Scope 2 is the carbon "embedded" in the electricity you buy from the grid. Even though you don't burn coal yourself, your DISCOM does on your behalf - and BRSR requires this to be disclosed.

```
Scope 2 (tCO2e) = Electricity kWh x 0.71 kg CO2e / kWh / 1000
```

Worked example

144,000 kWh x 0.71 = 102,240 kg CO_2e = 102.24 tCO_2e per year. This is the value displayed in Principle 6 parameter 2.1.b.

The 0.71 $kgCO_2e/kWh$ figure is India's national grid emission factor as published by the Central Electricity Authority (CEA) in its CO_2 Baseline Database for the Indian Power Sector, Version 21.0 (November 2025). Specifically, this is the Weighted Average Grid Emission Rate inclusive of renewables and captive generation, for FY 2024-25, all-India, excluding imports - the convention used for BRSR Scope 2 location-based reporting under the GHG Protocol.

v1.1 update (May 2026): Previous versions of this methodology used 0.82 $kgCO_2e/kWh$ from earlier CEA publications. The migration to V21.0 brings the factor in line with the latest published Indian grid mix, which has lower emissions intensity due to growing renewable share.

5.3 Scope 1 + 2 emissions intensity

```
Emission intensity = (Scope 1 + Scope 2) tCO2e / Turnover (Rs)
```

Same intensity convention as energy. Scope 3 (value-chain emissions) is offered as an optional input but the tool deliberately does not estimate Scope 3 for MSMEs - the data quality for supply-chain emissions in this segment does not yet support meaningful estimation.

6. Water withdrawal: kL per year

Water is the input where Ecosetu most explicitly relies on estimation, because MSMEs almost never have flow meters on borewells or pipelines. The tool offers users two paths.

6.1 Path A: User knows their water usage

If the user has a number - from a borewell pump rating, tanker invoices, or a water audit - they enter it directly in kL. The tool does no further calculation. The number flows straight into the disclosure.

6.2 Path B: User does not know - estimate from headcount

If the user has no water data, the tool estimates total annual water withdrawal from a per-capita per-day benchmark, multiplied by workers and working days.

```
Water (kL/year) = Workers x 40 litres x Working days / 1000
```

Worked example

An MSME with 50 workers operating 300 days a year: 50 x 40 x 300 = 600,000 litres = 600 kL per year.

The 40 litres/person/day benchmark is the standard Indian office-and-light-industry per-capita consumption assumption. For heavy water-using sectors (textile dyeing, paper, food processing) this figure will understate consumption significantly - the tool warns the user and recommends switching to direct measurement.

6.3 Water from tanker counts (intermediate path)

Many Bhavnagar / Saurashtra MSMEs purchase tanker water. The tool offers a tanker-count input as a third option.

```
Water (kL/year) = Tankers/month x 5,000 L x 12 / 1000
```

Worked example

An MSME buying 8 tankers a month: 8 x 5,000 x 12 / 1000 = 480 kL per year.

7. Waste and the savings opportunity layer

Waste in Principle 6 is mostly a disclosure exercise rather than a calculation - the user enters waste tonnage by category (hazardous / non-hazardous / e-waste / plastic) and the tool aggregates and reports in metric tonnes. No estimation factor is applied.

Money-saved-if-you-reduce projections

Ecosetu adds an optional "green savings" panel that shows what a 10% or 20% reduction in electricity would mean for the company's bottom line. This is a motivational projection, not an emissions claim, and is presented in rupees only.

```
Savings (Rs) = Current kWh x Target % reduction x Rs 8 per unit
```

We include this because MSME owners respond to rupee figures faster than to tCO2e figures. Once they engage with sustainability as a cost-saving exercise, the compliance follows easily.

8. Sources and references

Emission factors

- Diesel and LPG combustion factors: IPCC 2006 Guidelines for National Greenhouse Gas Inventories, Volume 2 (Energy), Chapter 2 (Stationary Combustion).
- Indian grid emission factor: Central Electricity Authority (CEA), CO2 Baseline Database for the Indian Power Sector, Version 21.0, November 2025. Specifically, the Weighted Average Grid Emission Rate inclusive of renewables and captive generation, FY 2024-25, all-India, excluding imports = 0.7117 tCO2/MWh (used in Ecosetu as 0.71 kgCO2e/kWh).
- Cross-referenced against: Bureau of Energy Efficiency (BEE) PAT scheme baselines and CDP India methodology notes for BRSR Scope 2 location-based reporting convention.

Energy conversions

- kWh to MJ and GJ: SI conversion (1 kWh = 3.6 MJ exactly).
- Diesel calorific value (9.9 kWh/L): derived from gross calorific value of approximately 35.6 MJ/L per IPCC and IEA reference values.

Water benchmarks

- Per-capita per-day consumption (40 L): aligned with Indian standard per-building water demand for offices and light industry. Reviewers comparing to IS 1172:1993 will note that standard prescribes 45 LPCD for offices with bathrooms; Ecosetu uses the slightly more conservative 40 LPCD figure.
- Standard tanker volume (5,000 L): industry-standard medium tanker capacity in western India.

BRSR alignment

- Disclosure parameters and reporting structure: SEBI Master Circular on Business Responsibility and Sustainability Reporting (current version), Section C, Principle 6.
- Intensity ratio conventions: consistent with BRSR-Core leadership indicators.

9. What Ecosetu deliberately does not do

It is just as important for a reviewer to know the boundaries of the tool. Ecosetu does not, and will not:

- Estimate Scope 3 (value-chain) emissions for MSMEs. The data quality does not yet support it. Scope 3 is offered as a user-entered optional input only.
- Apply sector-specific water multipliers. Heavy water sectors (textiles, paper, dyeing, food processing) must be flagged for direct measurement.
- Replace third-party assurance. The tool produces a BRSR-ready report; it does not opine on assurance, audit, or limited-review readiness.
- Translate to other frameworks (GRI, TCFD, ESRS) automatically. Mapping helpers may be added in future releases, with cross-walks made explicit.

Within those boundaries, every number in a Principle 6 report can be reconstructed line-by-line using only this document, the four to six operational inputs provided by the user, and a calculator.

10. Roadmap: from national defaults to state and time contextualisation

The current Principle 6 module uses national-average factors throughout. The next material upgrade to Ecosetu's calculation engine is contextualisation - so that an MSME in Tamil Nadu, Maharashtra or West Bengal sees factors and benchmarks specific to its state where these meaningfully differ:

- State-level electricity tariff defaults (replacing the Rs 8/unit national assumption).
- Regional grid emission factors where CEA publishes them by region.
- User-editable tanker volume and per-capita water inputs for sectors and contexts where the defaults do not fit.
- Annual version-pinning to CEA baseline releases, so reviewers can see exactly which CEA version applied for any given report.

Architecturally, this is a state-selector input feeding a lookup table - the formulas themselves do not change. Each release will be accompanied by a transparent versioned reference table.

Closing note from the founder

Ecosetu is built in public, under a proprietary license but with the source readable, precisely because trust in environmental disclosures cannot survive opacity. If you find a factor that should be updated, an edge case the tool handles poorly, or a sector where the estimation logic breaks down - please tell me. The number we ship together is more useful than the number I ship alone.

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